

F1 Tyre, Weather and Race Performance

Business Intelligence 1 - Assignment 1 - Summer semester 2026 - Group 9

OpenF1

Open-Meteo

Tableau

Star schema CSVs

Project focus

Lap-level analysis of tyre degradation, track/weather context, pit-window effects, driver consistency and circuit sensitivity.

Fact grain: one completed lap by one driver in one F1 race session.

Scope: completed Formula 1 Race sessions from 2023-2025.

Repository link: <https://github.com/woullufy/f1-weather-bi>

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Assignment flow and project coverage

The presentation follows the assignment tasks from data selection to insights.

1

Topic & data selection

F1 race performance, OpenF1,
Open-Meteo, circuit coordinates

2

BI justification

Five analytical questions and
race-strategy value

3

Data modelling

3NF source models and BI star schema

4

Data processing

Python pipeline transforms raw APIs
into Tableau-ready CSVs

5

Data analytics

Tableau dashboard components
mapped to Q1-Q5

6

Key insights

Management-style findings and
limitations

Appendix slides contain the deeper modelling, API and metric details for Q&A and grading review.

Project motivation and BI perspective

We frame the project as a race-strategy BI application.

Business / strategy problem

- F1 race performance is affected by tyre choice, tyre age, track temperature, weather, pit timing, driver pace and circuit characteristics.
- The problem is not a lack of data, but turning several time-series sources into comparable performance indicators.

BI framing

- The dashboard supports analytical comparison across dimensions.
- It is descriptive and diagnostic: what happened, where differences are visible, and where strategy assumptions may need closer inspection.

Interpretation boundary

- The project explores associations, not causal proof.
- Lap times also depend on fuel load, traffic, safety cars, car performance, team orders and driver behaviour.

Data Sources & Compatibility

OpenF1 provides performance facts; Open-Meteo adds external weather context.

Dataset	Type	Granularity	Role in project
OpenF1 API	Public F1 API	session, driver-lap, stint, pit, weather timestamp	main performance, tyre, pit and track-weather data
Open-Meteo	Public weather API	hourly weather by lat/lon/date	external ambient weather context
circuit_coordinates.csv	reference CSV	one row per circuit	bridge from F1 circuits to Open-Meteo coordinates

Compatibility key

Race session →
circuit key/location →
latitude/longitude →
Open-Meteo historical weather
window.

Performance facts

Driver-lap observations come from
OpenF1 laps, stints, drivers, teams
and track-side weather.

External context

Open-Meteo is not track
temperature; it is used as broader
ambient weather context.

Analytical questions and BI value

The questions cover tyre, weather, team, driver and circuit dimensions.

ID	Analytical question	Concrete BI value
Q1	Which tyre compounds lose lap-time performance fastest?	Compare compound degradation and stint behaviour
Q2	Does higher track temperature make soft tyres worse?	Assess soft-tyre performance under temperature stress
Q3	Which teams gain or lose the most around pit windows?	Compare pit-window execution and post-pit pace effects
Q4	Which drivers have the lowest lap-time variability?	Identify stable race pace after filtering abnormal laps
Q5	Are some circuits more sensitive to tyre wear or weather?	Support circuit-specific strategy preparation

All five questions are represented in the Tableau dashboard.

Fact-table grain, measures and dimensions

The dimensional model is built around a lap-level fact table.

Fact grain

One row = one completed lap by one driver in one F1 race session.

Why lap-level?

Tyre degradation and driver consistency require lap-by-lap observations.

Track temperature and pit windows also need timing close to individual laps.

A driver-race fact table would be too coarse for Q1, Q2 and Q4.

Measures

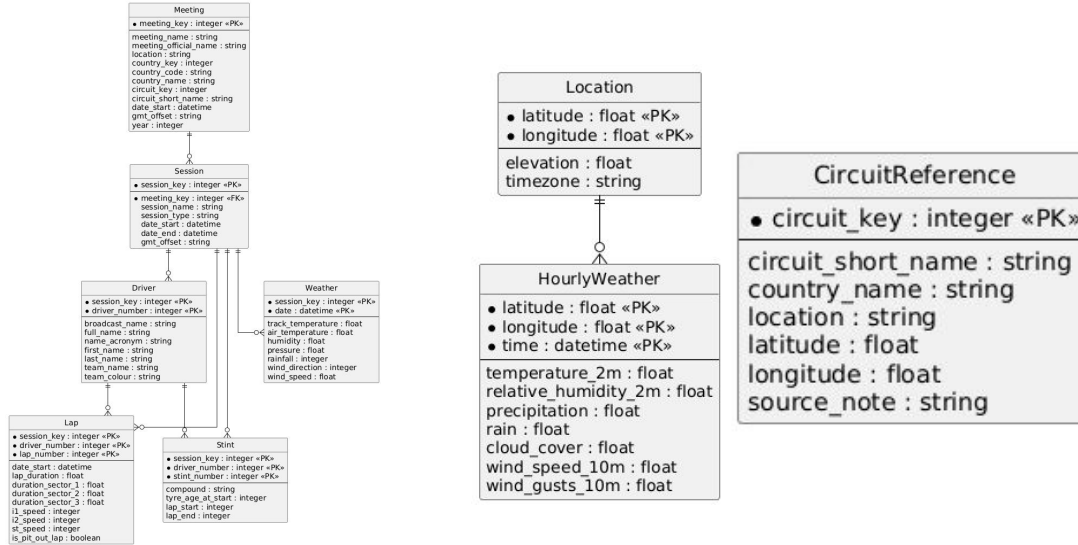
- lap duration
- sector durations
- tyre age
- lap-time deltas
- track temperature
- air temperature
- rainfall flag
- pit-window flags
- pit-window phase and gain/loss metrics

Dimensions

- Date
- Race / Grand Prix
- Circuit
- Driver
- Team
- Tyre stint
- Weather context

Source data modelling: operational-style entities

Each source dataset is documented and modelled before the BI star schema.



Operational entities: Weather, HourlyWeather, and CircuitReference. These schemas define the incoming raw data structure before transformation into the analytical star schema.

Design point

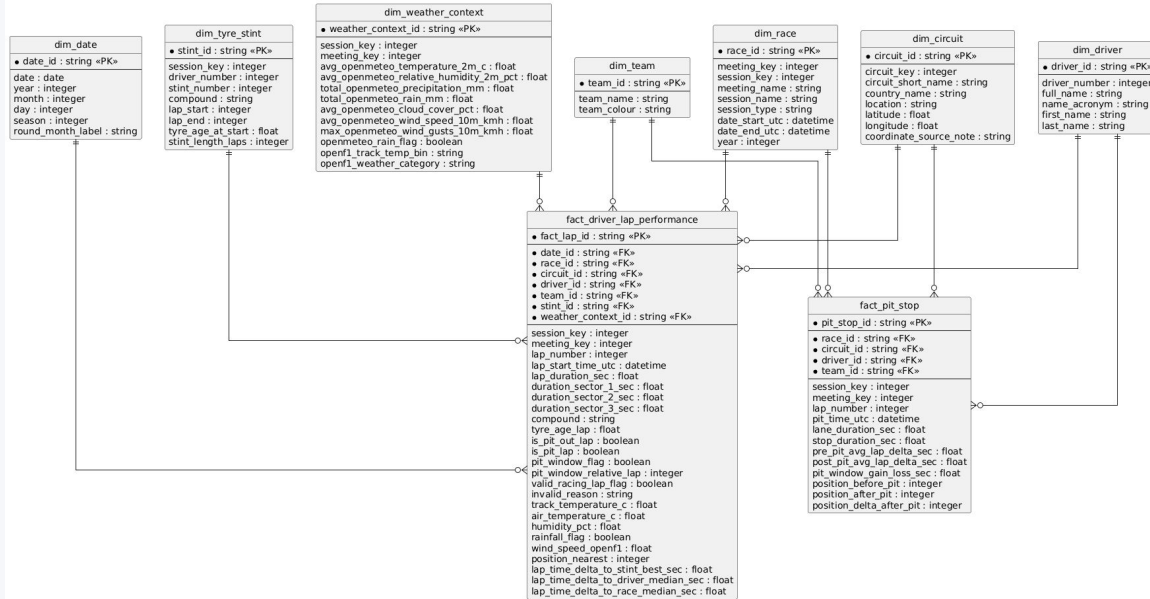
OpenF1 is modelled around sessions, drivers, laps, stints and weather observations.

Open-Meteo is modelled as hourly weather observations for locations.

The circuit reference bridges both worlds.

BI data model: star schema

The target model is optimized for Tableau queries and dashboard filters.



Central fact table

fact_driver_lap_performance

- One record per driver-lap
- Links to all dimensions

Main dimensions

date, race, circuit, driver, team, tyre
stint, weather context

Secondary fact table

fact_pit_stop handles the different grain of pit-stop events separately from lap-level performance.

BI data model: star schema

The target model is optimized for Tableau queries and dashboard filters.

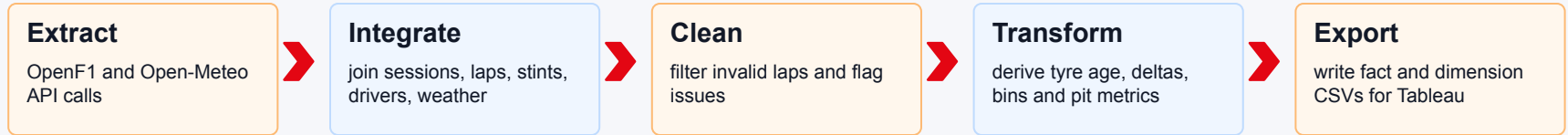
Main BI star schema: one central fact table

fact_pit_stop.csv is kept as an auxiliary output for pit-stop inspection, not as a second fact in the main schema.



Data processing pipeline

Python scripts transform raw API data into the target star schema.



Transformation examples

- derive tyre_age_lap
- calculate lap-time deltas
- assign temperature bins
- generate valid racing lap flags
- build surrogate IDs for dimensions

Integration challenges

- different time granularities
- nearest weather timestamp per lap
- circuit coordinate lookup
- pit-stop grain differs from lap grain

Cleaning examples

- missing lap times
- first laps
- pit laps and pit-out laps
- clear lap-time outliers
-

Data integration and quality decisions

The pipeline makes assumptions explicit and keeps problematic records traceable.

Integration choices

- session_key and meeting_key connect OpenF1 tables
- driver_number is session-specific
- laps join to stints by lap range
- OpenF1 weather joins by nearest previous timestamp
- Open-Meteo joins by circuit coordinates and date window

Quality controls

- -valid_racing_lap_flag separates laps suitable for pace analysis from laps excluded from clean metrics
- -pit-stop events are exported separately for inspection, while dashboard pit-window analysis uses lap-level pit-window phase fields
- circuit coordinates documented in reference file

Reusable output

- The processed CSVs can be regenerated from raw data and loaded into Tableau without manual post-processing.
- Manual input is limited to the stable circuit coordinate reference file.

Processed data coverage

The updated repository now contains a full processed dataset for the selected scope.

70

race sessions

77k

driver-lap facts

72k

valid racing laps

24

circuits

32

drivers

13

teams

3,948

tyre stints

2,339

pit stops

Processed CSVs

fact_driver_lap_performance.csv
fact_pit_stop.csv
dim_date.csv
dim_race.csv
dim_circuit.csv
dim_driver.csv
dim_team.csv
dim_tyre_stint.csv
dim_weather_context.csv

Weather and tyre coverage

Compounds: HARD, MEDIUM, SOFT, INTERMEDIATE, WET.

Weather context includes OpenF1 track weather and Open-Meteo ambient weather.

OpenF1 weather categories: 39 rain-flag sessions, 31 dry sessions.

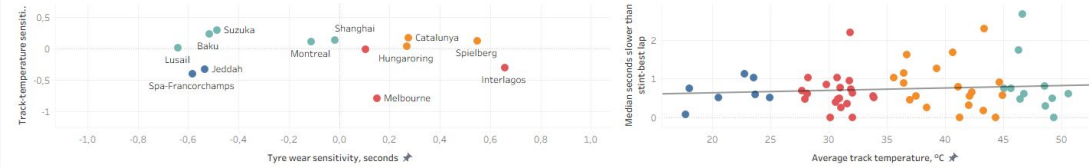
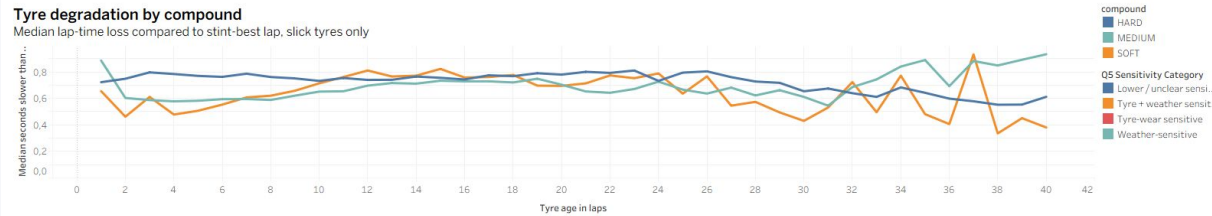
Dashboard design

The Tableau workbook gives one dashboard section per analytical question.

Formula 1 Tyre, Weather, and Lap Performance Dashboard

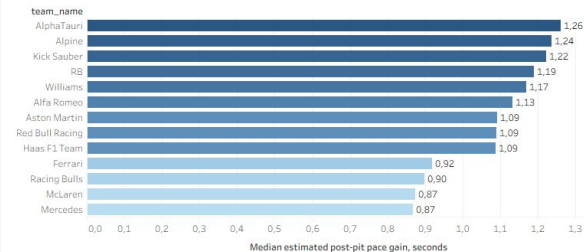
Tyre degradation by compound

Median lap-time loss compared to stint-best lap, click tyres only



Pit-window pace gain by team

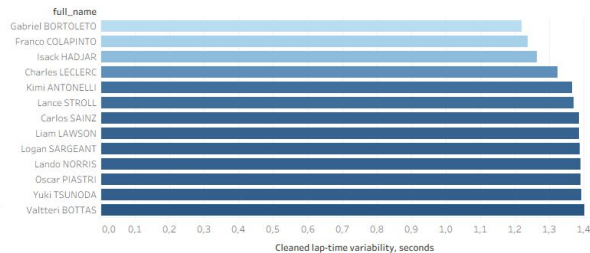
Median pace change from lap before pit to first normal lap after pit



Most consistent drivers

Top drivers by cleaned lap-time variability

Lower cleaned lap-time variability means more consistent race pace



Dashboard goal

Support comparison and drill-down across tyre, weather, team, driver and circuit dimensions.

Dashboard controls

Compound selection, valid-lap filtering, sensitivity categories and view-specific filters

Design rule

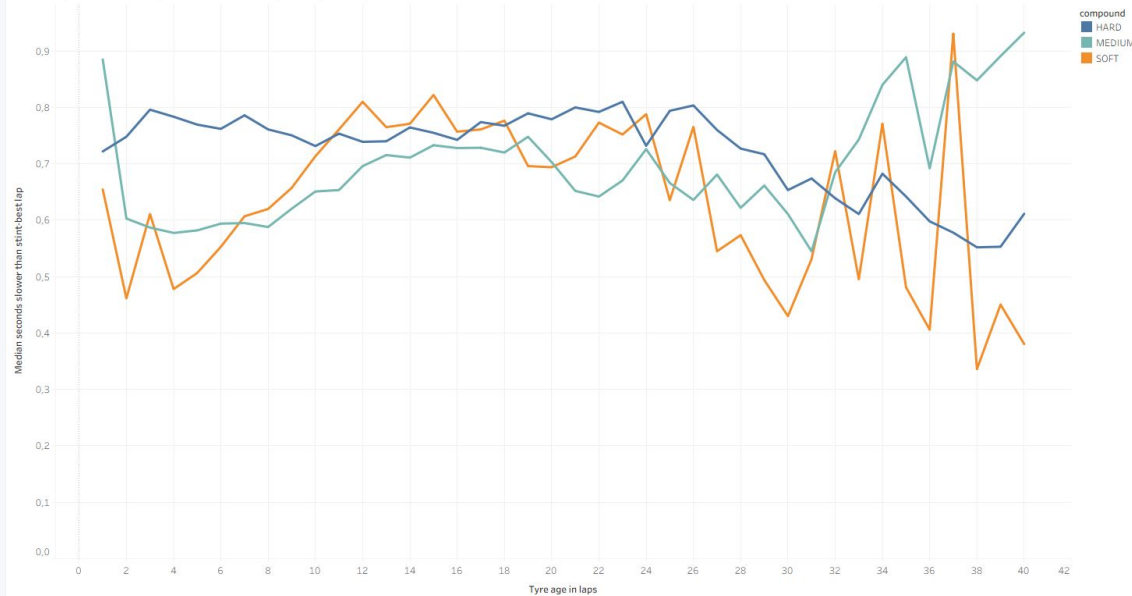
No unrelated charts: each view is tied to one of Q1-Q5.

Q1 - Tyre degradation

Which tyre compounds lose lap-time performance fastest?

Tyre degradation by compound

Median lap-time loss compared to stint-best lap, slick tyres only



The trend of median of Clean Lap Delta for tyre_age_lap. Color shows details about compound. The data is filtered on valid_racing_lap_flag and sum of Clean Lap Delta. The valid_racing_lap_flag filter keeps True. The sum of Clean Lap Delta filter includes everything. The view is filtered on compound, which keeps HARD, MEDIUM and SOFT.

How it is measured

Median lap-time loss compared to the best valid lap in the same stint, grouped by tyre age and compound.

Observation

The compound lines are visible but noisy. Aggregated tyre effects differ by compound and by stint age, especially for longer stints.

Interpretation

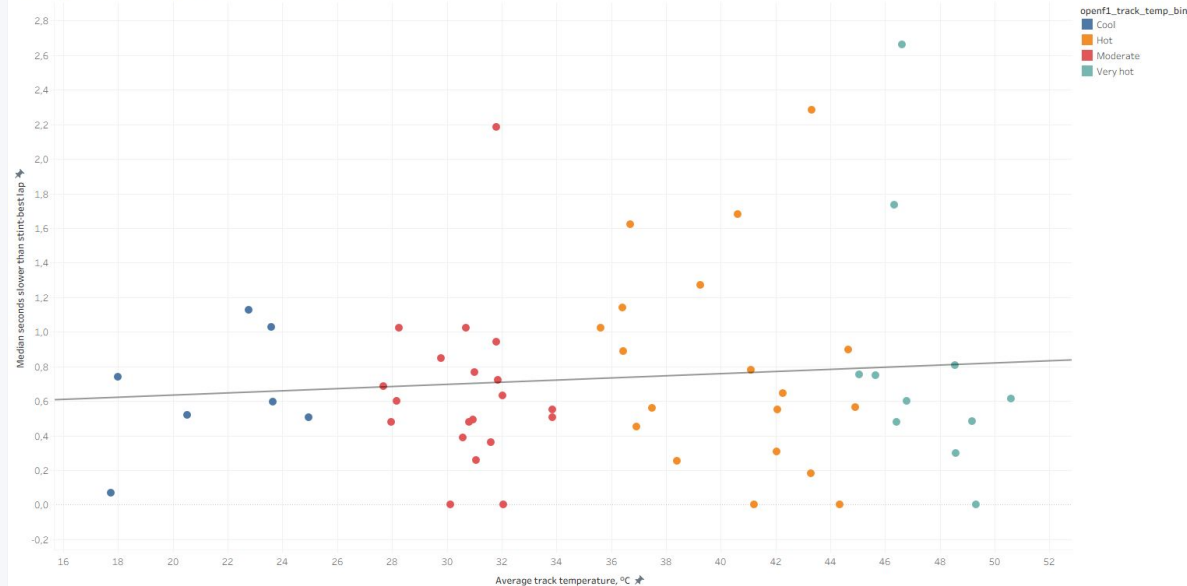
This view supports compound comparison, but it should not be treated as pure physical tyre degradation because fuel load, traffic and strategy also affect lap time.

Q2 - Track temperature and soft tyres

Does higher track temperature make soft tyres worse?

Soft tyre performance by track temperature

Median lap-time loss for soft tyre laps across race sessions



The plot of median of Clean Lap Delta for track_temperature_c. Color shows details about openf1_track_temp_bin. Details are shown for meeting_name and circuit_short_name (dim_circuit.csv). The data is filtered on valid_racing_lap_flag and compound. The valid_racing_lap_flag filter keeps True. The compound filter keeps SGT.

How it is measured

SOFT tyre laps are compared by average track temperature and OpenF1 temperature bins.

Observation

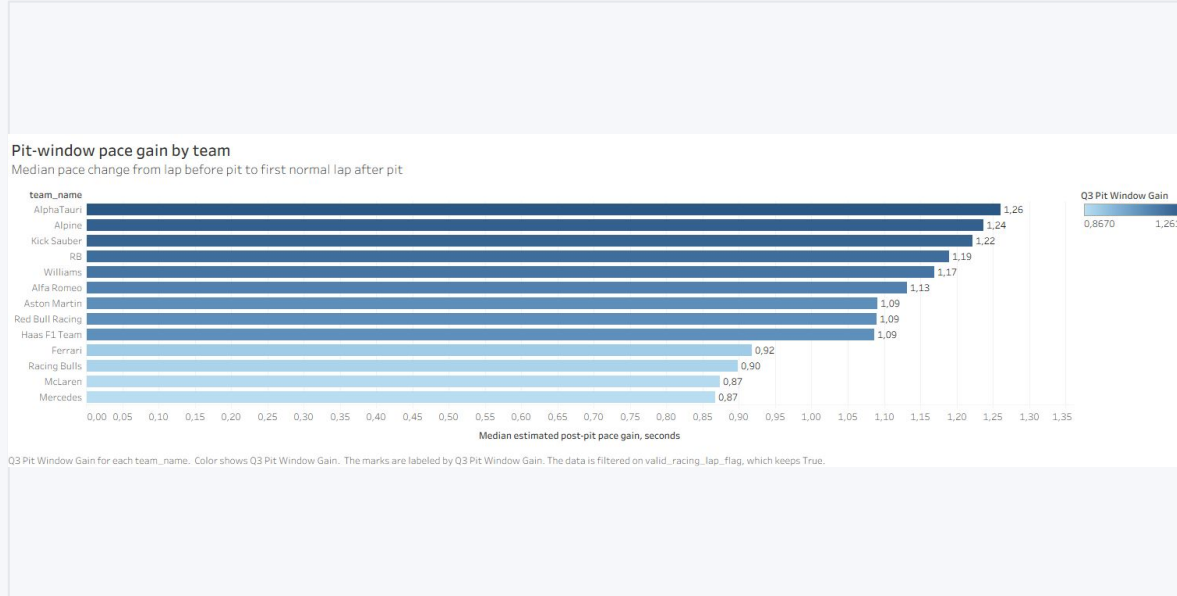
The current data shows a weak positive trend line, but the relationship is scattered rather than clearly linear.

BI value

The chart helps identify sessions where temperature should be considered when evaluating soft-tyre behaviour.

Q3 - Pit-window strategy

Which teams gain or lose the most around pit windows?



How it is measured

Pit-window effects are analysed inside the main lap-level fact table using pit-window phase fields. The metric compares the lap before a pit stop with the first normal lap after the pit stop.

Observation

The dashboard ranks teams by estimated post-pit pace gain. Top-ranked teams in the current view include AlphaTauri, Alpine and Kick Sauber.

Caution

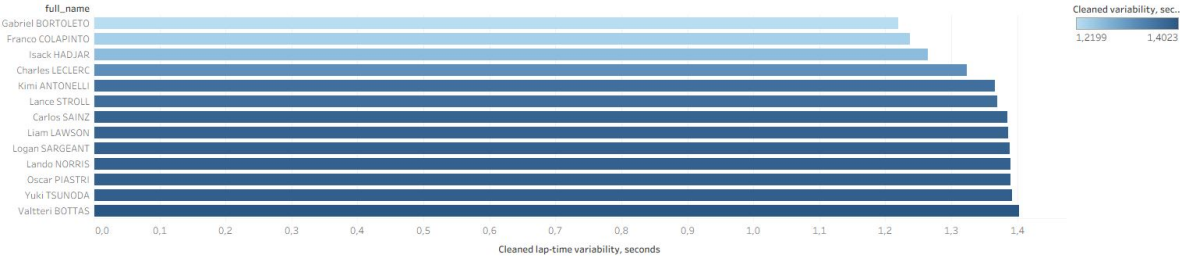
This is a pit-window pace metric, not a full causal strategy model. Traffic, safety cars and opponent strategy still matter.

Q4 - Driver consistency

Which drivers have the lowest lap-time variability?

Most consistent drivers

Top drivers by cleaned lap-time variability
Lower cleaned lap-time variability means more consistent race pace



Q4 Driver Lap Variability Clean for each full_name. Color shows Q4 Driver Lap Variability Clean. The view is filtered on Q4 Valid Lap Count and full_name. The Q4 Valid Lap Count filter includes values greater than or equal to 500. The full_name filter keeps 13 members.

How it is measured

Cleaned lap-time variability is calculated after filtering non-racing laps and abnormal observations.

Observation

Lower bars indicate more consistent race pace. The dashboard shows a focused ranking after applying valid-lap and abnormal-lap filters.

BI value

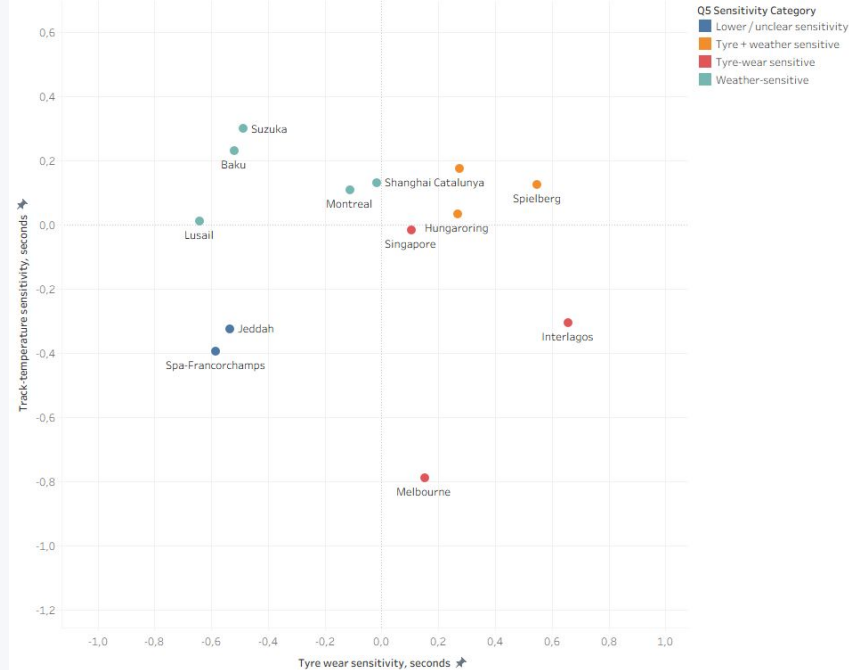
The view separates consistency from final race result, which helps compare driver race pace stability.

Q5 - Circuit effects

Are some circuits more sensitive to tyre wear or weather?

Circuit sensitivity to tyre wear and weather

Tyre-age effect vs. track-temperature effect by circuit



QS Tyre Wear Sensitivity vs. QS Weather Sensitivity. Color shows details about QS Sensitivity Category. The marks are labeled by circuit_short_name (dim_circuit.csv). Details are shown for circuit_short_name (dim_circuit.csv).

How it is measured

Circuit-level tyre-wear sensitivity compares old-tyre vs young-tyre lap-time loss. Track-temperature sensitivity compares hot/very-hot contexts against cooler contexts.

Observation

Circuits separate into categories: lower/unclear sensitivity, tyre-sensitive, weather-sensitive, and combined tyre + weather sensitivity.

BI value

This supports circuit-specific strategy preparation and helps identify where further drill-down is needed.

Key insights for management

Dashboard findings should be presented as decision-support insights, not causal proof.

1. Tyres

Tyre compound and tyre age are useful dimensions for explaining lap-time loss, but the aggregate trend is noisy and race-dependent.

2. Temperature

Soft tyre performance appears more variable as track temperature rises; the relationship is not purely linear.

3. Pit windows

The pit-window view creates a comparable team-level post-pit pace-gain metric, but it should be interpreted as a simplified strategy proxy rather than pure pit-crew performance.

4. Drivers

Driver consistency can be compared independently of final positions when abnormal laps are filtered.

5. Circuits

Circuit sensitivity differs, making track-specific preparation necessary for tyre and weather strategy.

Main takeaway

The BI pipeline turns scattered race and weather data into a reusable basis for strategy-oriented comparisons.

Limitations and interpretation boundaries

The project provides analytical insight, not a complete race simulation.

Method limitations

- association analysis, not causal proof
- lap times are affected by fuel load, traffic, safety cars, team orders and car performance
- weather data has different granularities
- Open-Meteo is ambient/reanalysis context, not track surface temperature
- safety-car/red-flag effects are not fully controlled

Data and dashboard limitations

- some categories have fewer observations than others
- pit-window gain/loss is a simplified metric
- wet and extreme-weather samples may be small
- insights should be validated by drill-down into specific races and circuits

Conclusion and repository contents

The project covers the full BI flow required by Assignment 1.

Project delivery

- public data sources
- 3NF source models
- star schema
- automated Python pipeline
- processed CSVs
- Tableau workbook
- dashboard screenshots

What the dashboard supports

- compound comparison
- weather/temperature exploration
- pit-window comparison
- driver consistency ranking
- circuit sensitivity analysis

Repository files

- README.md
- docs/
- src/
- data/processed/
- tableau/
- dashboard_screenshots/
- slides/

Repository link: <https://github.com/woullufy/f1-weather-bi>

Appendix

Additional details for Q&A and grading review

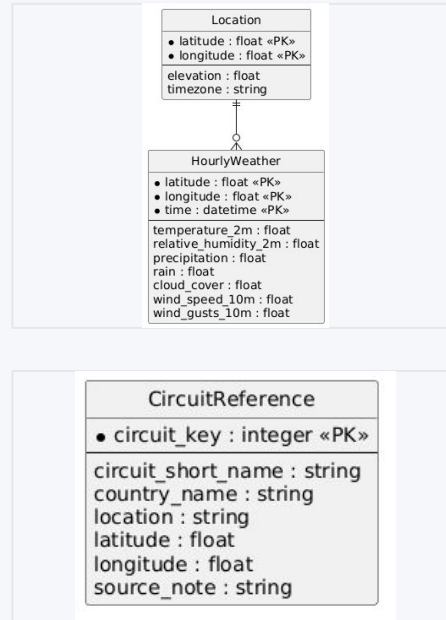
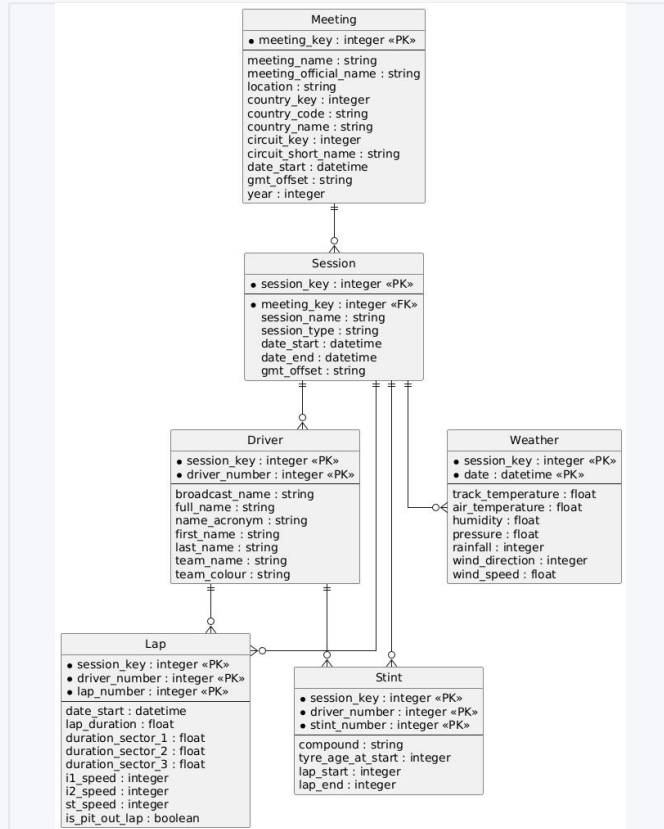
Appendix A - OpenF1 endpoints used

The project uses only endpoints needed for Q1-Q5.

Endpoint	Purpose	Priority
/sessions	race session filter and time window	MVP
/meetings	event and circuit metadata	MVP
/drivers	driver and team dimensions	MVP
/laps	main lap fact data	MVP
/stints	compound and tyre age	MVP
/weather	track temperature and track-side weather	MVP
/pit	pit-window fact table	MVP-plus
/position	optional position before/after pit stop	optional
/race_control	optional safety-car/red-flag filtering	optional

Appendix B - Source data models

Operational source models before BI transformation.

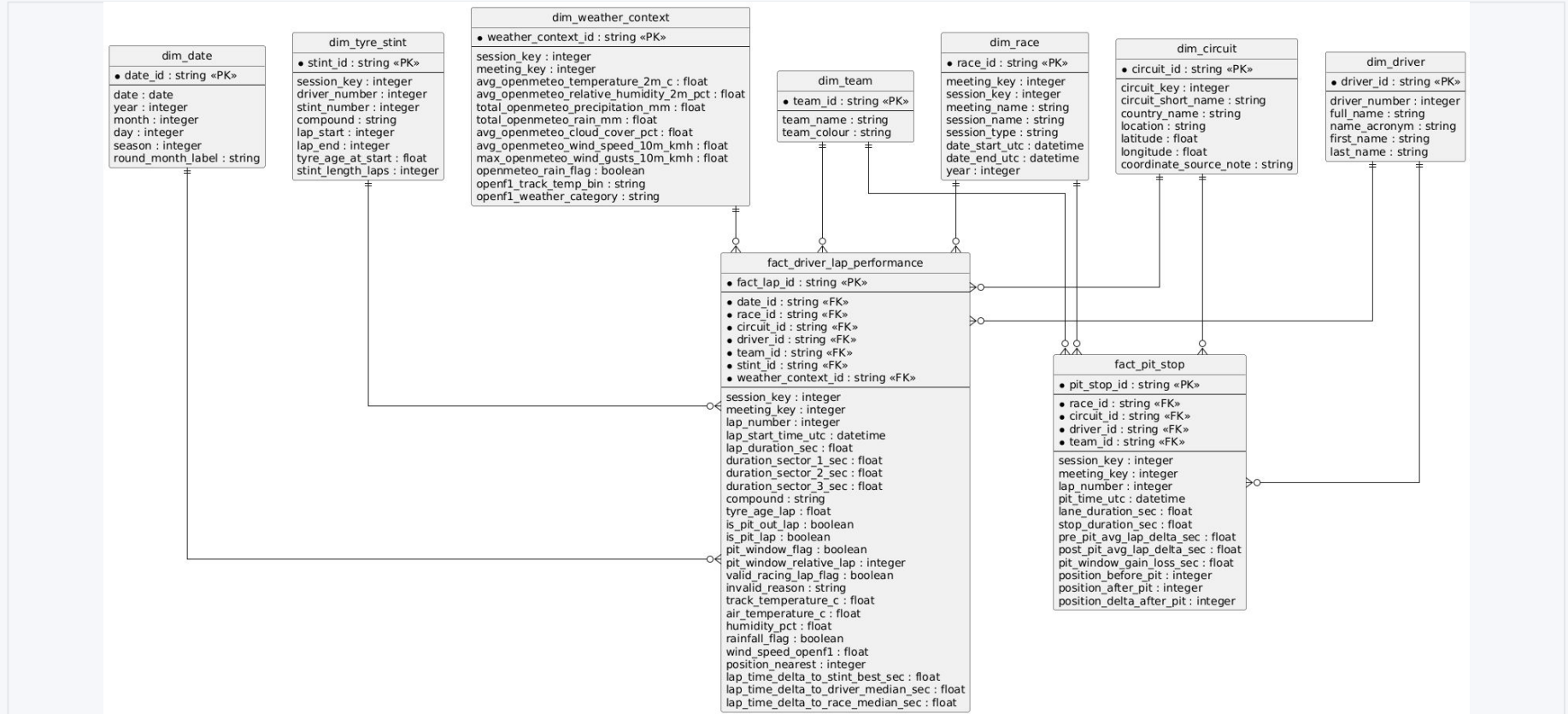


ER modelling note

OpenF1 and Open-Meteo are first described as source systems. The final dashboard does not directly use the operational ER structure; the pipeline transforms it into the star schema.

Appendix C - Star schema detail

Target BI schema used for Tableau.



Appendix D - Metric definitions and cleaning logic

Derived fields are created in the Python pipeline.

Metric / flag	Definition / purpose
tyre_age_lap	tyre_age_at_start + (lap_number - lap_start)
lap_time_delta_to_stint_best_sec	lap_duration_sec - best valid lap in same driver-stint
lap_time_delta_to_driver_median_sec	lap_duration_sec - median valid driver-race lap
degradation_slope	slope of normalized lap-time loss over tyre age
driver_lap_variability	standard deviation of cleaned normalized lap-time deltas
pit_window_gain_loss_sec	pre-pit normalized pace minus post-pit normalized pace
valid_racing_lap_flag	true only for laps suitable for pace/degradation analysis

Invalid reasons include missing lap time, first lap, pit lap, pit-out lap, pit-window lap, outlier lap and optional safety-car/red-flag laps.

Appendix E - Repository and deliverables

Files included in the updated project repository.

Area	Files / folders
Raw data	data/raw/openf1, data/raw/openmeteo
Processed data	data/processed/*.csv
Reference data	data/reference/circuit_coordinates.csv
Pipeline	src/fetch_openf1.py, src/fetch_openmeteo.py, src/process_pipeline.py
Documentation	README.md, docs/project_handoff.md, docs/source_*.md, docs/star_schema.md
Tableau	tableau/f1_tyre_weather_dashboard.twbx
Screenshots	dashboard_screenshots/*.png

Final submission: PDF slides with repository link via Moodle, plus repository containing source links, transformed data, pipeline and Tableau workbook.